Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Hyderabad, India

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Summary

AI/ML Engineer with two years at AWS Professional Services shipping production ML systems. Trained dense neural network on 121K+ LeetCode contest records to MAE ∼7.8. Architected medical image classification pipeline on SageMaker with blue-green endpoints, auto-scaling, and drift detection. Built LSTM stock predictor and neural network framework from scratch in NumPy. Upstream contributor to Feast feature store, Apache Airflow, and Prefect. 5x AWS certified including **AWS Machine Learning Engineer – Associate** and **AI Practitioner**. LeetCode Guardian, 2166 peak.

Experience

Amazon Web Services (AWS) Aug 2024 – Present
ProServe (Cloud Consultant) - DevOps/MLOps Hyderabad, India

- **MLOps Pipeline – SageMaker (SME Program, Jun-Aug 2025):** Built the SageMaker pipeline that trains, gates and ships medical image classification models, with blue-green endpoint deployment, auto-scaling, and PSI drift detection that triggers retraining – zero-downtime updates with automatic rollback on error and latency alarms. Full IaC via Terraform + CodePipeline; API Gateway + Lambda inference API behind CloudFront frontend. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0).
- **Lead DevOps Consultant – RWS (Feb 2026 – Present):** Cut per-account setup from 3+ days to under 4 hours (∼90%) building the AWS foundation for a data-centre exit and a second landing zone entirely in Terraform: Organizations OUs with Control Tower baselines, SCPs, tag policies, and least-privilege account baselines. Delivered two Transit Gateway hubs with Network Firewall inspection and VPN/Direct Connect, plus self-service network onboarding. Used Kiro and Claude Code for requirement analysis, specs, and design documentation.
- **DevOps Consultant – State Street (Oct 2024 – Aug 2025):** Built Terraform for about 10 AWS services with automated tests (tftest + PyTest) on the regulated landing zone team, part of the path that opened 38+ services to application teams. Designed Amazon Detective enablement via an AFT customization. Project closed with a **10/10 CSAT**.
- **DevOps Consultant – DTCC (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC patterns on high-sensitivity financial workloads. Used Amazon Q to accelerate code analysis and migration assessments.
- **Internal Impact:** Published a second AWS sample, Terraform for AWS Organizations governance on Control Tower (aws-samples, MIT-0), covering six policy types, Control Tower controls, and delegated admin for 15 services. MLOps SME Program Graduate (2025). Published APG pattern *Secure PR Previews and Approval-Gated Releases for Amplify Gen 2* (plus 2 more peer-reviewed in publication pipeline). 5x TFC Ambassador (AI/ML, NGDE Graduated, CloudOps, Agentic AI, Serverless). Associate Speaker Certified.

Amazon Web Services (AWS) Mar 2024 – Jul 2024
Associate Professional Services – DevOps (Internship) Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – zero-downtime releases. Established observability with X-Ray and CloudWatch distributed tracing.

Ikarus-3D May 2023 – Jul 2023
Software Developer Associate (Internship) Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD, and deployed SonarQube on EC2 + RDS to embed continuous code quality analysis into the delivery pipeline.

Industry Certifications (5)

AWS: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
AWS: AI Practitioner | Cloud Practitioner

Education

National Institute of Technology Warangal, Telangana Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

Sanskar College of Professional Studies (DAVV University), Indore May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

Projects

LeetCode Rating Predictor Dec 2023
Python, TensorFlow, FastAPI, React, GraphQL GitHub | Live
◇ Trained dense neural network on **121K+ contest records** reaching **MAE** ∼**7.8** for post-contest rating predictions; feature engineering pipeline pulls raw contest data via LeetCode GraphQL API.
◇ Served the TensorFlow model behind a FastAPI backend with a React frontend for real-time user predictions.

GroundZeroNN Oct 2023
Python, NumPy, Neural Networks from Scratch GitHub
◇ Built a neural network framework in pure Python/NumPy – forward/backward propagation, gradient descent, activation functions, and batch normalization implemented from scratch without any ML library.

StockSage AI Nov 2023
Python, TensorFlow, LSTM, Streamlit GitHub
◇ LSTM stock price predictor combining technical indicators (RSI, MACD, Bollinger Bands) with sequential price data; interactive Streamlit dashboard for real-time predictions and historical back-tests.

Bedrock Multi-Model MCP Server 2026
Python, Amazon Bedrock, MCP Protocol GitHub
◇ MCP server routing tool calls across 500+ Amazon Bedrock foundation models through a single endpoint – Claude, Llama, Mistral, Titan – with per-model streaming and cost telemetry.

Claude Cost Optimizer 2026
TypeScript, Anthropic API, React Live
◇ Analyzes Claude usage traces to recommend model + prompt-caching strategies that cut spend 30–60% without accuracy loss.

Other Notable Projects: GitScope (Chrome extension, GitHub analytics), Ledger Sync (personal-finance reconciler), Instagram Likes Leaderboard.

Open Source Contributions

(11 merged upstream)

- **axios** (109K☆) – Synthesize AxiosError message from AggregateError on dual-stack connection failures (#11059, merged)
- **apache/airflow** (44K☆) – Add `template_fields` to SalesforceBulkOperator (#63109, merged)
- **PrefectHQ/prefect** (21K☆) – Document custom deployment steps as Python functions (#20956, merged)
- **feast-dev/feast** (9K☆) – Claude Code agent skills for Feast feature store (#6081, merged)
- **awslabs/agent-plugins** – Guard defusedxml extension load (#132, merged)
- **awslabs/mcp** – Fix Kendra documentation menu to match the source directory (#2607, merged)
- **expressjs/multer** – Stop files skipped by fileFilter from consuming maxCount (#1426, merged)
- **lucide** (24K☆) – Vue Icon kebab-case icon-node prop fix (#4512, merged)
- **aws-ia/terraform-aws-vpc** – Cloud WAN attachment-replacement root cause; fix co-authored into v4.8.0 (merged)
- **hashicorp/terraform-provider-aws** – IoT CloudWatch metric substitution validation (#46867, open)

Technical Skills

ML & Deep Learning: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas; Dense Networks, CNNs, LSTMs, Ensemble Methods, Feature Engineering, Model Evaluation (MAE/RMSE, precision/recall)
MLOps on AWS: SageMaker (Training, Pipelines, Endpoints, Model Monitor, Drift Detection), Blue-Green Deployments, Model Versioning, CodePipeline, Feast Feature Store
GenAI & LLMs: Amazon Bedrock, RAG Pipelines, LLM APIs (Anthropic, OpenAI), MCP Protocol, Prompt Engineering, Claude Code, Amazon Q, Kiro
Serving & Infrastructure: FastAPI, Lambda, API Gateway, CloudFront, Docker, Terraform, GitHub Actions, GraphQL
Languages: Python, SQL, TypeScript, JavaScript, HCL, C++, Java

Achievements

- **MLOps SME Program Graduate** (AWS, 2025). Published APG pattern: *Secure PR Previews and Approval-Gated Releases for Amplify Gen 2* (plus 2 more peer-reviewed in publication pipeline).
- **LeetCode Guardian** – 2166 peak rating, 1300+ problems solved, 107 contests. **GeeksforGeeks** 500+ DSA problems. **HackerRank** 6-Star Problem Solving, 5-Star C++.
- **Rank 1289 (Top 5%)** – Google Kick Start Round E '22. *Certificate*
- **1st Place** – Coding Cauldron '23 (NIT Trichy); QFiesta IGNOSI '22 (NIT Warangal).